AVERRA

AVERRA — Biomedical Technology

# Practical Portfolio

Engineering better movement intelligence.

| | |
|---|---|
| **Course** | Entrepreneurship for Engineers (SEC-ENT-301), VJTI Mumbai |
| **Faculty** | Krupali Prakashkumar Nakar |
| **Founders** | Pranshu Kumar · Pawan Shinde |
| **Products** | ParkinSense (wearable wellness and movement intelligence) · TremoSense (assistive active-stabilisation spoon) |
| **Contents** | 15 mandatory models (M1.1–M5.3), references, evidence appendix, checklists |
| **Submission date** | 16 Oct 2026 |

Evidence labels used throughout: **Sourced** (published or official figure, cited), **Model** (from the AVERRA financial model), **Estimate** (founders' planning assumption) and **Simulated** (semester case-study data used to complete an activity where real data is still being collected; to be replaced with real records).

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M1 · Activity 1 · Mandatory Model M1.1

# Customer-Discovery Evidence Pack

| | | | |
|---|---|---|---|
| **Module** 1 | **Activity** 1 | **Date completed** 12 Aug 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · simulated interview case study | | | |
| *Data note: Interview records are a simulated, anonymised case study; the secondary evidence in section 5 is published and cited.* | | | |

## 1. Target customer and problem hypothesis

**TremoSense:** Adults in urban India living with Parkinson's disease or essential tremor whose hand tremor makes eating difficult, and the family members who help them at meals, regularly lose independence and dignity at mealtimes and currently rely on help from others, adapted cutlery or avoiding certain foods.

**ParkinSense:** Wearable users, including people living with tremor and their families, want continuous insight into how they move between clinic visits, but today their movement is only observed for a few minutes during a consultation.

## 2. Mom Test interview questions (past behaviour only)

1. Tell me about the last meal where your (or your family member's) tremor made eating harder. What happened?
2. What did you do about it at that meal? What do you usually do?
3. What have you already tried for eating with tremor (spoons, weighted cutlery, help from someone, changing food)? How did that go?
4. Have you spent money on anything to help with tremor or eating? What was it, roughly how much, and where did you hear about it?
5. How often in the last week did tremor change what or where you ate?
6. When did you last discuss tremor with a neurologist or therapist? What did they measure, and what happened next?
7. Do you currently track symptoms, activity or sleep in any way (watch, app, diary)? Show me the last thing you recorded.
8. Who else is involved when a decision about buying a health device is made in your home?

Rules followed: no pitching before the questions, no "would you buy", no leading questions; record exact words; ask for specifics (last time, how often, how much).

## 3. Interview evidence log (minimum 5 real interviews)

| # | Date | Respondent (anonymised) | Exact words (simulated) | Pain · frequency · workaround · spend |
|---|---|---|---|---|
| 1 | 22 Aug 2026 | Daughter, 52, caring for father with PD (Dadar) | “At lunch he spills half the dal, so I just feed him now. He hates it. Last Sunday he refused to eat with the family.” | Spills daily · feeds him herself · ≈1 h/day of her time |
| 2 | 24 Aug 2026 | Man, 67, essential tremor (Thane) | “I stopped ordering soup at restaurants about two years ago. I hold the cup with both hands at home.” | Avoids liquids in public · weekly · two-hand grip · no spend |
| 3 | 27 Aug 2026 | Woman, 71, PD (Sion) | “My son bought a heavy steel spoon online for ₹1,800. It helped a bit but my wrist pains after ten minutes.” | Daily · weighted spoon · ₹1,800 spent |
| 4 | 29 Aug 2026 | Son, 38, father with PD (Matunga) | “The neurologist sees him for ten minutes every three months. Whatever happened in between, we have to remember it.” | Clinic snapshot · quarterly · keeps WhatsApp notes |
| 5 | 2 Sep 2026 | Occupational therapist (Bandra clinic) | “About one in five of my tremor patients asks about adaptive cutlery. The imported ones cost more than most families can spend.” | Recurring ask · weekly · recommends built-up handles |
| 6 | 4 Sep 2026 | Wife, 64, husband with ET (Wadala) | “He eats before guests arrive so nobody sees him shake. We tried a ₹600 rubber grip; it didn't stop the shaking.” | Embarrassment · each visit · ₹600 grip |

## 4. Pattern analysis

| Signal | Count out of N = 6 | Evidence (interview #) |
|---|---|---|
| Same pain mentioned (eating disrupted) | 5 / 6 | 1, 2, 3, 6 (and 5 as a clinician) |
| Weekly or more frequent | 5 / 6 | 1, 2, 3, 5, 6 |
| Has an existing workaround | 6 / 6 | All |
| Has already paid money/time/reputation | 4 / 6 | 1 (time), 3 (₹1,800), 6 (₹600), 2 (avoids eating out) |
| Clinician measures movement only in clinic | 2 / 6 | 4, 5 |

## 5. Secondary evidence already in hand (labelled as published, not interviews)

| Who | Published statement | Source |
|---|---|---|
| Clinicians | “Obtaining only a snapshot of symptoms during a single clinical office visit does not provide enough time resolution to determine how to optimize symptomatic benefit.” | Hadley, Riley & Heldman, Digital Biomarkers 2021 https://karger.com/dib/article/5/3/206/823097/Real-World-Evidence-for-a-Smartwatch-Based |
| People with tremor | “I am embarrassed by my tremor when I try to eat or drink in public.” | Most-endorsed item among people with essential tremor; 63.2% found their tremor embarrassing. Kellner et al., J Neurol Sci 2017 https://pmc.ncbi.nlm.nih.gov/articles/PMC5739080 |
| Caregivers (India) | “Out of 100 caregivers, 73 reported to have "severe" caregiver burden.” | Parkinson's caregivers, Southern India. Kurupath, Arathil & Bansal, JCDR 2020 https://jcdr.net/ReadXMLFile.aspx?id=14384 |

## 6. Continue / kill decision

Kill trigger set before the interviews: stop TremoSense as a consumer product if fewer than 3 of 5+ interviewees describe tremor disrupting meals at least weekly, or none has spent money or effort on a workaround.

**Decision: CONTINUE.** 5 of 6 describe tremor disrupting meals at least weekly and 4 of 6 have already spent money or time on a workaround that does not fully work, which clears our kill trigger. The pattern matches the published evidence in section 5.

M1 · Activity 2 · Mandatory Model M1.2

# Opportunity Scorecard & "Why Now?" Memo

| | | | |
|---|---|---|---|
| **Module** 1 | **Activity** 2 | **Date completed** 19 Aug 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · sourced and model figures | | | |
| *Data note: Market figures are sourced and computed in the financial model; percentage filters are labelled estimates.* | | | |

## 1. Bottom-up TAM, SAM and SOM (India, revenue per year)

All inputs are on the *Market* and *Assumptions* sheets of the financial model. Revenue per customer spreads the device price over a 3-year life, net of GST.

| Step | TremoSense | ParkinSense (PD households, beachhead) |
|---|---|---|
| Population base | Indians aged 60+ = 14.9 Cr (UNFPA 2023) | Same base |
| Patients | PD 308.9 per lakh aged 60+ → 460,261; ET 89.96 per lakh → 1,276,575 | PD aged 60+ = 460,261 |
| Customers with the need | (PD + ET) × 50% eating-disruptive tremor = 868,418 | 460,261 households |
| ₹ per customer per year | ₹9,999 ÷ 1.05 GST ÷ 3 yrs = ₹3,174 | ₹7,999 ÷ 1.18 ÷ 3 + 40% × ₹169 × 12 = ₹3,069 |
| **TAM** | **868,418 × ₹3,174 = ₹275.66 Cr** | **460,261 × ₹3,069 = ₹141.26 Cr** |
| **SAM** (urban 31.1% × under neurologist care 40%) | **108,031 people = ₹34.29 Cr** | **57,256 = ₹17.57 Cr** |
| **SOM** (Year 3, what our channels can reach) | **30 clinics × 4/month × 12 + 50 online/month × 12 = 2,040 = ₹64.76 L** (1.9% of SAM) | **30 clinics × 2/month × 12 + 60 online/month × 12 = 1,440 = ₹44.19 L** (2.5% of SAM) |

**Company totals:** TAM ₹416.92 Cr · SAM ₹51.86 Cr · Year-3 SOM ₹108.95 L. Customer counts are not summed (one household may buy both).

**Upside, not in plan totals — ParkinSense as a general wearable:** IDC India 2024: 3.5 Cr smartwatches × ₹3,069 = ₹10742 Cr TAM; × 2.8% premium share = SAM ₹300.77 Cr; SOM 0.25% / 0.5% / 1% of SAM buyers = ₹75.19 L / ₹1.50 Cr / ₹3.01 Cr.

> Estimates to state openly: 50% eating-disruptive share, 40% under neurologist care, 3-year device life, 40% subscription take-up, clinic and online volumes. ET prevalence comes from a single rural-Gujarat study. Year-3 SOM is a target, not data.

## 2. "Why Now?" memo

**The shift.** Movement sensing on the wrist became mainstream and regulator-accepted, and India created a defined approval route for devices like ours. Dated evidence: Apple opened a Movement Disorder API in ResearchKit (June 2018); Rune Labs' Apple Watch Parkinson's monitoring received FDA clearance (June 2022); licensing became mandatory for all medical devices in India from 1 October 2023; UNFPA projects India's 60+ population to rise from 14.9 Cr to 34.7 Cr by 2050 (September 2023); BMJ projects 25.2 million people with Parkinson's worldwide by 2050 (March 2025). Hardware cost collapsed too: the microcontroller with an on-board 6-axis IMU used in both products lists at US$15.90 (Seeed Studio).

**Why it was harder three years ago.** Before October 2023 India had no mandatory, clearly defined licensing route for devices like ours, so a product could not be planned and costed end to end, and wrist-based PD monitoring had no regulatory precedent before 2022.

**Why the window will not stay open.** Global wearable brands (WHOOP, Garmin, Google, Amazfit) already sense movement; if they add tremor features, or imported active spoons (₹35,980–₹58,480 in India today) become cheaper, the low-cost Indian gap closes.

## 3. 90-second pitch (≈150 words; rehearse aloud, no notes)

> About 4.6 lakh Indians over sixty live with Parkinson's, and more have essential tremor. Their movement is measured for a few minutes in a clinic, and at home a 4–12 hertz tremor turns a spoon into a spill. Published studies say a clinic visit is "only a snapshot", and people with tremor are most embarrassed eating in public. AVERRA builds two products on one sensing stack. TremoSense is an active stabilising spoon: our current build rejects about 90 percent of tremor on a bench simulator, founder-reported. ParkinSense is a ₹7,999 wellness band with tremor detection. Bottom-up, our

> serviceable market is ₹51.9 Cr a year, and through 30 clinics plus online we target ₹109 L by year three. Next, we run a preregistered user study with at least ten people, and continue only if median tremor reduction reaches 50 percent.

Word count ≈ 150. Timed delivery: 86 s (third rehearsal, 9 Sep 2026).

## 4. Opportunity scorecard (1 = weak, 5 = strong; every score cites its evidence)

| Criterion | Score | Evidence behind the score |
|---|---|---|
| Size of the problem | 4 | Bottom-up TAM ₹416.9 Cr; PD and ET prevalence from published studies (Cureus 2025; Mov Disord 2021) |
| Pain intensity | 4 | Most-endorsed ET item is embarrassment eating in public (Kellner 2017); 73/100 Indian caregivers report severe burden (JCDR 2020) |
| Willingness to pay | 2 | Not yet measured; imported spoons sell at ₹35,980+ and GyroGlove at ≈₹5.6 L, but no AVERRA price test yet |
| Timing (why now) | 4 | Dated regulatory and technology shifts 2018–2023 (memo above) |
| Competition / differentiation | 3 | No screenless band lists tremor detection; active spoons exist but cost 4–6× our planned price |
| Team fit | 4 | Founders built both prototypes; two papers; bench results |
| Evidence from own customers | 1 | Own interviews pending (see M1.1) |
| **Total** | **22 / 35** | Strong on problem and timing; weakest on own-customer and pricing evidence |

M1 · Activity 3 · Mandatory Model M1.3

# Venture Money-Model & First-Budget Pack

| | | | |
|---|---|---|---|
| **Module** 1 | **Activity** 3 | **Date completed** 26 Aug 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · model figures | | | |
| *Data note: Budget lines come from the financial model; personal savings and expenses are illustrative.* | | | |

## 1. The money model on one page

| What we sell | Who pays | How much | How often | Cost to deliver one unit |
|---|---|---|---|---|
| TremoSense active stabilising spoon (sale; rental option) | Family of the person with tremor, referred by a clinic or OT centre | ₹9,999 incl. 5% GST (planned); ₹8,999 from M19; rental ₹799/month | Once per ≈3-year device life | ₹4,830 small-batch COGS + 15% clinic margin + 5% warranty + ₹250 shipping → contribution ₹2,538 |
| ParkinSense wellness band + optional insights plan | The user, online | ₹7,999 incl. 18% GST + optional ₹199/month | Device once; plan monthly (40% take-up assumed) | ₹3,530 COGS + 10% online fees + warranty + shipping → contribution ₹1,982 |
| Research-use (RUO) kits | University labs | ₹74,999 per kit | One-off | ₹22,650 → contribution ₹40,658 |

**Revenue-model choice, defended:** a one-time device sale (with rental for families who cannot pay upfront) fits an assistive product bought once by a family, and keeps every unit profitable from the first sale. A subscription is offered only on ParkinSense, where ongoing insights create ongoing value; it is optional, so it never blocks access.

**Brand positioning (Digital Marketing I):** For people who want to understand how they move, and for people whose tremor makes eating hard, AVERRA is a biomedical wearables company that turns precise movement sensing into everyday insight (ParkinSense) and active assistance (TremoSense), at prices built for India. **Organic channels:** neurologist and OT referrals, Parkinson's and ET support groups, LinkedIn and YouTube demo videos of the bench rig, the AVERRA website.

## 2. First three months (M1–M3, Jul–Sep 2027), zero-based from the financial model

Each line is justified from zero (what it buys), not copied forward. Source: Forecast sheet, rows for M1–M3.

### One-time costs

| Line | M1 | M2 | M3 | 3-month total |
|---|---|---|---|---|
| Industrial design contractor | ₹45,000 | ₹45,000 | ₹0 | ₹90,000 |
| Legal, accounting, admin | ₹25,000 | ₹12,000 | ₹12,000 | ₹49,000 |
| Patent: attorney + filing fees | ₹0 | ₹55,000 | ₹0 | ₹55,000 |
| TremoSense study units (12 × Scenario A cost) | ₹0 | ₹0 | ₹58,341 | ₹58,341 |

### Monthly recurring costs

| Line | M1 | M2 | M3 | 3-month total |
|---|---|---|---|---|
| Founder stipends (2×₹15k Phase 1 · 2×₹30k M7–24 · 2×₹45k Y3) | ₹30,000 | ₹30,000 | ₹30,000 | ₹90,000 |
| Prototyping materials | ₹25,000 | ₹25,000 | ₹25,000 | ₹75,000 |
| Tools, cloud, software | ₹8,000 | ₹8,000 | ₹8,000 | ₹24,000 |

| | M1 | M2 | M3 | 3-month total |
|---|---|---|---|---|
| Subtotal | ₹1,33,000 | ₹1,75,000 | ₹1,33,341 | ₹4,41,341 |
| **Contingency 10%** | ₹13,300 | ₹17,500 | ₹13,334 | ₹44,134 |
| **Total 3-month budget** | **₹1,46,300** | **₹1,92,500** | **₹1,46,676** | **₹4,85,476** |

## 3. Personal runway

| Founder | Personal savings (₹) | Personal monthly expenses (₹) | Personal runway (months) |
|---|---|---|---|
| Pranshu Kumar | ₹60,000 | ₹8,000 | 60,000 ÷ 8,000 = 7.5 |
| Pawan Shinde | ₹45,000 | ₹7,500 | 45,000 ÷ 7,500 = 6.0 |

Context from the plan: each founder draws a ₹15,000/month stipend in Phase 1 (M1–M6), ₹30,000 in M7–M24 and ₹45,000 in Year 3 (Forecast sheet). Personal runway shows how long each founder can continue if stipends are deferred.

M2 · Activity 1 · Mandatory Model M2.1

# Evidence-Backed Business Model Canvas + Value Proposition Canvas

| | | | |
|---|---|---|---|
| **Module** 2 | **Activity** 1 | **Date completed** 2 Sep 2026 | **Venture** AVERRA |
| **Evidence status:** Completed | | | |
| *Data note: Each block is classed as evidence-backed or assumption.* | | | |

## 1. Business Model Canvas (all nine blocks)

| Block | AVERRA | Evidence-backed or assumption? |
|---|---|---|
| **Customer segments** | Everyday wearable users; households living with Parkinson's or essential tremor; research labs | Assumption (sizes from published prevalence) |
| **Value propositions** | ParkinSense: a wearable for everyone with built-in tremor detection. TremoSense: active tremor assistance at an Indian price | Partly evidenced (bench results; no user study) |
| **Channels** | Clinic and OT referral; direct online; institutional procurement | Assumption |
| **Customer relationships** | Clinic-assisted trial; caregiver onboarding; remote support | Assumption |
| **Revenue streams** | TremoSense sale ₹9,999 or rental ₹799/mo; ParkinSense ₹7,999 + optional ₹199/mo; RUO kits ₹74,999 | Assumption (prices planned, untested) |
| **Key resources** | Sensing and control know-how; two working prototypes; two papers; repositories | Evidence-backed (repos, papers) |
| **Key activities** | User studies; CDSCO and ISO 13485; firmware and DSP; small-batch assembly | Evidence-backed for engineering; assumption for regulatory route |
| **Key partners** | Neurology clinics; OT centres; university labs; a test lab; a contract assembler; ICMR (clinical evaluation in progress, founder-reported) | Partly evidenced (ICMR founder-reported; no clinic partners yet) |
| **Cost structure** | Phase 1 ₹8.76 L: team, study units, user study, IP. Then QMS, EMC, inventory and tooling, outreach | Evidence-backed for component prices (Seeed, servo retailers); rest estimated |

## 2. Value Proposition Canvas

### TremoSense — person with tremor and their family

| Customer profile | Value map |
|---|---|
| **Jobs:** eat independently; eat in public without embarrassment; reduce the help needed at every meal | **Product:** active two-axis stabilising spoon (IMU → EKF → LQR → two servos) |
| **Pains:** spills; embarrassment (Kellner 2017); imported active spoons cost ₹35,980–₹58,480; weighted spoons tire the hand | **Pain relievers:** ≈90% tremor rejection on the bench (founder-reported); planned price ₹9,999; built and serviced in India |
| **Gains:** dignity; fewer meals needing help; caregiver time back | **Gain creators:** rental option ₹799/month; clinic-assisted trial before buying |

### ParkinSense — wearable user (including people living with tremor)

| Customer profile | Value map |
|---|---|
| **Jobs:** understand activity, vitals and movement every day; share a record with a clinician | **Product:** screenless wrist band streaming IMU + PPG over BLE to a dashboard |
| **Pains:** clinic sees only a snapshot (Hadley 2021); screenless bands cost ₹12,000–₹25,990 and none lists tremor detection | **Pain relievers:** tremor score, frequency and burden; tremor never counted as steps; ₹7,999 with no mandatory subscription |
| **Gains:** long battery; a clear history to discuss with a doctor | **Gain creators:** live dashboard; optional ₹199/month insights plan |

## 3. The three riskiest assumptions and how we test each

| # | Riskiest assumption | Why it would sink us | Test (one line) |
|---|---|---|---|
| 1 | Bench tremor rejection turns into easier eating with real food and real hands | No product without it | Preregistered self-feeding study, n ≥ 10, M6: continue only if median reduction ≥50% |
| 2 | Families will pay ₹9,999 (or rent at ₹799/month) | Unit economics fail | In-clinic trial-to-purchase test, 10 clinics, 8 weeks: ≥30% buy at ₹9,999 |
| 3 | Clinicians will refer patients | Channel and CAC fail | Outreach to 10 neurologists/OT centres: ≥5 agree to host a trial; track referrals per clinic per month |

M2 · Activity 2 · Mandatory Model M2.2

# MVP Validation Experiment & Decision

| | | | |
|---|---|---|---|
| **Module** 2 | **Activity** 2 | **Date completed** 9 Sep 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · simulated semester pilot; funded study planned | | | |
| *Data note: Pilot results are simulated (n = 5); the preregistered n ≥ 10 study is planned for M6 of the funded plan.* | | | |

## 1. What exists today (evidence)

| MVP | What was run | Result so far | Status |
|---|---|---|---|
| TremoSense current build (Zephyr, 208 Hz, higher-torque servos) | Motorised tremor simulator at 4, 6, 8 Hz, ±5°; 2 people with tremor tried it informally | ≈90% tremor rejection on the bench (founder-reported; logs to be published); positive informal feedback, no metric | Bench evidence only |
| ParkinSense prototype + dashboard | Bench conditions (rest, 5 Hz fixture, walking, arm motion); BLE soak; power profiling | 8/8 bench conditions correct; ≈7 mA. Clinical evaluation with ICMR in progress, results pending (founder-reported) | Bench evidence; clinical results pending |

No success metric was preregistered before the bench tests above, so they are reported as bench evidence, not as a validated MVP. The protocol below is preregistered for the funded M6 study. A small semester pilot (n = 5, simulated case study) was run on the same protocol to check feasibility.

## 2. Preregistered experiment (planned for M6)

| Item | Protocol |
|---|---|
| MVP being tested | 12 TremoSense study units (Scenario A build) |
| Target user | Adults with Parkinson's or essential tremor whose tremor disrupts eating; recruited via neurologists/OT centres; ethics approval first |
| Hypothesis | Using TremoSense reduces tremor amplitude at the spoon head during a standard self-feeding task, and users value it enough to pay ₹9,999 |
| Success metric 1 | Median reduction in spoon-head tremor amplitude (IMU) vs a normal spoon, same participant, same task |
| Success metric 2 | Share of participants who say they would pay ₹9,999 (or rent at ₹799/month) after the session |
| Preregistered thresholds | Persevere: median reduction ≥50% AND ≥60% would pay · Redesign and re-test once: 30–50% · Kill TremoSense: <30% · Pivot to clinic rental: works but <60% would pay |
| Procedure | Each participant: 10 spoonfuls of a standard food with a normal spoon, then 10 with TremoSense (order randomised); log IMU on handle and head; short interview |
| Duration and sample | 4 weeks in M5–M6; n ≥ 10 |
| ParkinSense pass mark (M10) | Dominant-frequency error ≤5% vs a reference device; zero false steps during tremor |

## 3. Semester pilot result against the thresholds (simulated, n = 5)

| Metric | Preregistered threshold | Pilot result (simulated) | Met? |
|---|---|---|---|
| Median tremor reduction | ≥50% | 58% (range 41–67%) | Yes (indicative) |
| Would pay ₹9,999 or rent at ₹799/month | ≥60% | 3 of 5 = 60% (2 buy, 1 rent) | Yes, at the threshold |
| Participants completed | ≥10 | 5 of 5 | No — pilot sample only |

**Decision: PERSEVERE to the funded study.** Both thresholds were met in the pilot, but n = 5 is below the preregistered sample, so the result is indicative only; the kill / continue decision stays with the n ≥ 10 study at M6. Two of five found the handle heavy, which led to the change in M5.2.

M2 · Activity 3 · Mandatory Model M2.3

# Pricing, Margin & Funnel Performance Pack

| | | | |
|---|---|---|---|
| **Module** 2 | **Activity** 3 | **Date completed** 16 Sep 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · model pricing; simulated funnel | | | |
| *Data note: Prices and margins are from the model; funnel counts and deposits are simulated semester data.* | | | |

## 1. Two candidate prices for TremoSense

| | Price 1: ₹9,999 (value-based, launch) | Price 2: ₹8,999 (cost-plus after cost reduction) |
|---|---|---|
| Rationale | Lowest price with LTV:CAC above 3× at small-batch cost; ≈¼ of the cheapest imported active spoon (₹35,980) | Cost-plus once tooling and supplier quotes bring COGS to ₹2,720 (Scenario C, from M19) |
| Price ex-GST (÷ 1.05) | ₹9,523 | ₹8,570 at B · ₹8,570 at C |
| Clinic margin 15% | ₹1,428 | ₹1,286 at B · ₹1,286 at C |
| Net revenue | ₹8,094 | ₹7,285 at B · ₹7,285 at C |
| COGS | ₹4,830 | ₹4,830 at B · ₹2,720 at C |
| Warranty 5% | ₹476 | ₹429 at B · ₹429 at C |
| Fulfilment | ₹250 | ₹250 |
| **Contribution per unit** | **₹2,538 (31.4%)** | **₹1,776 (24.4%) at B · ₹3,886 (53.3%) at C** |

Contribution margin = selling price − all variable costs of one unit (GST is collected for the government and excluded). Source: Price Matrix and TremoSense Costs sheets. ParkinSense at ₹7,999: net ₹6,101 − COGS ₹3,530 − warranty − shipping = ₹1,982 (32.5%).

## 2. Funnel (AARRR) — assumed targets, not measured

| Stage | Plan target (model) | Plan per 1,000 reached | Conversion formula | Semester result (simulated) |
|---|---|---|---|---|
| Reach (Acquisition) | 1,000 caregivers via 10 clinics | 1,000 | — | 6,550 (Instagram 6,200 + support group 350) |
| Enquiry / trial (Activation) | 25% try it in clinic | 250 | trials ÷ reached | 34 enquiries (0.52%) → 5 pilot trials |
| Purchase (Revenue) | 40% of trials buy | 100 | buyers ÷ trials | 3 refundable deposits = 60% of trials |
| Retention | ≥70% daily use at day 30 | 70 | active at day 30 ÷ buyers | 4 of 5 pilot users still using at day 28 (80%) |
| Referral | 15% refer another family | 15 | referrers ÷ buyers | 1 of 3 depositors referred a family (33%) |

> **Preorders (simulated semester case study):** 3 refundable ₹1,000 deposits at ₹9,999 from pilot caregivers, 18–19 Sep 2026, total ₹3,000. Real payment records will replace this entry; deposits are refundable and are not counted as revenue in the model.

## 3. Plan to collect real willingness-to-pay evidence

1. During the M6 user study, ask each participant a Van Westendorp price ladder (too cheap / bargain / getting expensive / too expensive) and record exact answers.
2. Offer a refundable ₹1,000 preorder deposit at ₹9,999 to participants and caregivers after the session; count deposits, not compliments.
3. At two clinics, show ₹9,999 and at two others ₹8,999 (A/B by clinic) and record enquiry → trial → deposit conversion.

## 4. IP note (Legal/IP basics)

**Patent:** a provisional patent filing is *planned* for M2 (not filed) covering the stabilisation control arrangement. **Trademark:** "AVERRA", "ParkinSense" and "TremoSense" should be searched and filed (Class 10 medical devices, Class 9 software).

**Copyright:** firmware, dashboard code and documentation are protected automatically; record authorship and inventor assignments to the company before any filing.

M3 · Activity 1 · Mandatory Model M3.1

# Operating System & Delivery-Process Blueprint

| | | | |
|---|---|---|---|
| **Module** 3 | **Activity** 1 | **Date completed** 23 Sep 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · estimated timings | | | |
| *Data note: Step times are estimates for a 100-unit pilot batch, to be measured on the first batch.* | | | |

All times below are **estimates** for a 100-unit pilot batch; none has been measured yet. Measure each step on the first pilot batch and replace the estimate.

## 1. TremoSense: order to support

| # | Step | Owner | Estimated time |
|---|---|---|---|
| 1 | Clinic/OT referral → in-clinic trial booked | Clinic + AVERRA | 1–3 days |
| 2 | Trial session and order (sale or rental) | Clinic staff | 30 min |
| 3 | Pick sub-assemblies from stock | Operations | 10 min |
| 4 | Final assembly (gimbal, servos, battery) | Contract assembler | 40 min |
| 5 | **Servo calibration + tremor-rejection test on the simulator rig** | Founders (hardware & test) | **45 min per unit** |
| 6 | Safety check, cleaning of food-contact head, packing | Operations | 15 min |
| 7 | Courier to home or clinic | Courier | 2–4 days |
| 8 | Onboarding call with caregiver; 7-day check-in | Support | 20 min + 10 min |

**Bottleneck:** step 5. One rig and two founders give ≈9 units/day, slower than assembly. **Redesign (no new spend first):** run calibration as a scripted batch on the existing rig (firmware self-test + automatic pass/fail on the logged IMU), test 4 units per rig cycle, and only hand-check failures. Target ≈12 min per unit.

## 2. ParkinSense: online order to support

| # | Step | Estimated time |
|---|---|---|
| 1 | Online order and payment | instant |
| 2 | Flash firmware, pair, register device ID | 10 min |
| 3 | **BLE streaming soak test (packet loss, battery)** | **8 h per batch** |
| 4 | Strap fit, pack | 10 min |
| 5 | Courier | 2–4 days |
| 6 | App/dashboard onboarding email + support | 15 min |

**Bottleneck:** the soak test. **Redesign:** a 20-slot charging-and-soak rack that tests a full batch overnight in parallel.

## 3. SOP — TremoSense final calibration and test (new hire can follow)

1. Scan the unit's serial QR code; the test script opens a log file named with the serial number.
2. Charge to ≥80% (green LED). If not reached in 60 min, tag “battery” and set aside.
3. Mount the handle in the simulator clamp, spoon head pointing up; tighten until the handle does not slip.
4. Run *self-test*: the servos sweep ±20°; check there is no grinding and both axes return to level.
5. Run the simulator at 4, 6 and 8 Hz for 20 s each; the script computes tremor rejection from the handle and head IMUs.
6. Pass if rejection ≥ the release threshold at every frequency (threshold set from the published bench logs); the script prints PASS/FAIL.
7. On PASS, wipe the food-contact head with food-safe sanitiser, fit the cap and pack with the manual. On FAIL, tag the failure code and send to rework.
8. Upload the log; the order is released for dispatch only when a PASS log exists.

## 4. Lean wastes and how we remove them

| Waste | Where it shows up | How we reduce it |
|---|---|---|
| Waiting | Assembled units wait for calibration | Batch self-test script (above); calibrate as units arrive, not at the end |
| Inventory | Pilot batch of 100 units bought before demand is proven | Buy servos and enclosures in 25-unit lots tied to clinic orders; keep only sub-assemblies |
| Defects | Servo or boost-converter failures found by the customer | 100% end-of-line test with a stored PASS log; track failure codes to fix root causes |

M3 · Activity 2 · Mandatory Model M3.2

# Unit-Economics & Runway Model

| | | | |
|---|---|---|---|
| **Module** 3 | **Activity** 2 | **Date completed** 26 Sep 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · model forecast | | | |
| *Data note: All figures are forecasts from the financial model; AVERRA has no actual revenue yet.* | | | |

Every number is a forecast from the financial model (Unit Economics and Forecast sheets). AVERRA has no actual revenue yet; nothing below is an actual financial result.

## 1. Unit economics

| Metric | Formula | TremoSense | ParkinSense |
|---|---|---|---|
| CAC | planned acquisition spend ÷ customers acquired | ₹1,079 | ₹1,500 |
| Contribution per unit | price ex-GST − channel − COGS − warranty − shipping | ₹2,538 | ₹1,982 |
| LTV | lifetime contribution per customer | ₹2,538 × 1.4 purchases (40% repurchase) = ₹3,553 | ₹1,982 + 40% × ₹129/mo × 20 mo = ₹3,011 |
| LTV : CAC | LTV ÷ CAC | 3.29× | 2.01× |
| CAC payback | CAC ÷ contribution per sale | 0.43 of the first sale → paid back on the first sale | 0.76 of the first sale → first sale |

## 2. 12-month cash forecast (Jul 2027 – Jun 2028), funded by the ₹10.51 L round only

| Month | Cash out | Cash in | Net cash flow | Cumulative | Closing cash |
|---|---|---|---|---|---|
| M1 · Jul-27 | ₹1,46,300 | ₹0 | -₹1,46,300 | -₹1,46,300 | ₹9,04,871 |
| M2 · Aug-27 | ₹1,92,500 | ₹0 | -₹1,92,500 | -₹3,38,800 | ₹7,12,371 |
| M3 · Sep-27 | ₹1,46,676 | ₹0 | -₹1,46,676 | -₹4,85,476 | ₹5,65,695 |
| M4 · Oct-27 | ₹1,59,500 | ₹0 | -₹1,59,500 | -₹6,44,976 | ₹4,06,195 |
| M5 · Nov-27 | ₹1,15,500 | ₹0 | -₹1,15,500 | -₹7,60,476 | ₹2,90,695 |
| M6 · Dec-27 | ₹1,15,500 | ₹0 | -₹1,15,500 | -₹8,75,976 | ₹1,75,195 |
| M7 · Jan-28 | ₹2,80,500 | ₹0 | -₹2,80,500 | -₹11,56,476 | -₹1,05,305 |
| M8 · Feb-28 | ₹1,87,000 | ₹0 | -₹1,87,000 | -₹13,43,476 | -₹2,92,305 |
| M9 · Mar-28 | ₹4,23,500 | ₹0 | -₹4,23,500 | -₹17,66,976 | -₹7,15,805 |
| M10 · Apr-28 | ₹4,42,915 | ₹58,308 | -₹3,84,607 | -₹21,51,582 | -₹11,00,411 |
| M11 · May-28 | ₹6,68,841 | ₹0 | -₹6,68,841 | -₹28,20,423 | -₹17,69,252 |
| M12 · Jun-28 | ₹2,77,915 | ₹1,68,833 | -₹1,09,082 | -₹29,29,505 | -₹18,78,334 |

Closing cash = ₹10.51 L raised − cumulative net outflow (Funding Scenarios sheet, Scenario C: Round 1 only, no grant). Negative closing cash means tranche 2 must already have closed.

## 3. Burn, runway and zero-cash date

- Phase-1 burn (M1–M6) = ₹8,75,976 ÷ 6 = **₹1,45,996/month** (no cash in).
- Year-1 gross burn = ₹31,56,646 ÷ 12 = **₹2,63,054/month**; net burn after ₹2,27,141 cash in = **₹2,44,125/month**.
- Runway on this round = ₹10,51,171 ÷ ₹1,45,996 = **7.2 months** at Phase-1 burn.
- **Projected zero-cash date: January 2028 (M7)**, when Phase-2 spending starts. Tranche 2 (₹37.71 L) must close by then; if it is late, hiring, inventory and testing pause and the buffer stretches to about M14.

## 4. Bad-growth scenario (toggle: price cut to grow faster)

| Year-2 TremoSense | Normal plan | Bad growth |
|---|---|---|
| Price | ₹9,999 | ₹5,999 (discount to chase volume) |
| Units sold | 695 | 1,390 (twice the volume) |
| Contribution per unit (Scenario B cost) | ₹2,538 | -₹510 |

| Year-2 TremoSense | Normal plan | Bad growth |
|---|---|---|
| CAC per customer | ₹1,079 | ₹1,079 |
| Contribution − CAC per customer | ₹1,459 | -₹1,589 |
| **Total effect on cash** | **₹10,14,005** | **-₹22,08,710** |

In the bad-growth case every extra customer loses ₹1,589, so doubling volume deepens the loss by ₹32.23 L versus the plan and pulls the zero-cash date earlier. This is why the plan holds ₹9,999 until a verified cost reduction (Scenario C) makes ₹8,999 profitable. (Illustrative toggle built from the model's Price Matrix; not a separate sheet in the workbook.)

M3 · Activity 3 · Mandatory Model M3.3

# Legal-Structure, GST & Compliance Blueprint

| | | | |
|---|---|---|---|
| **Module** 3 | **Activity** 3 | **Date completed** 30 Sep 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · illustrative GST working | | | |
| *Data note: GST figures are worked examples; rates and HSN codes to be confirmed with a chartered accountant.* | | | |

> Educational blueprint, not legal or tax advice. GST rates and HSN codes for TremoSense (assumed 5%) and ParkinSense (assumed 18%) must be confirmed with a chartered accountant.

## 1. Structure comparison for AVERRA

| | Proprietorship | Partnership | LLP | Private Limited |
|---|---|---|---|---|
| Liability | Unlimited, personal | Unlimited, joint | Limited | Limited |
| Paperwork | Lowest | Low (deed) | Moderate (annual LLP filings) | Highest (ROC filings, audit, board meetings) |
| Raising equity | Not possible | Not practical | No equity shares; VCs rarely invest | Equity shares, SAFEs/CCPS, ESOPs; standard for investors |
| DPIIT / Startup India / SISFS | No | Registered firms eligible | Eligible | Eligible |
| Fit for a two-founder medical-device startup | No | No | Partial | **Best fit** |

**Recommendation: Private Limited.** We plan to raise equity in two tranches and grant ESOPs, which needs shares; limited liability matters for a device used on the body; and DPIIT recognition plus the Startup India Seed Fund application (applied for, not secured) require a recognised entity.

## 2. Mock GST calculation (hypothetical example, per unit)

| | TremoSense @ ₹9,999 (5% assumed) | ParkinSense @ ₹7,999 (18% assumed) |
|---|---|---|
| Output GST charged to customer | ₹9,999 − ₹9,999/1.05 = ₹476 | ₹7,999 − ₹7,999/1.18 = ₹1,220 |
| Input tax credit on purchases (18% on bought-in parts) | (COGS ₹4,830 − labour/QC ₹600) × 18% = ₹761 | COGS ₹3,530 × 18% = ₹635 |
| **Net GST payable** | **-₹285** → credit (inverted duty: input rate > output rate; refund can be claimed under Sec 54(3), CGST Act) | **₹585** payable |
| For 100 units | -₹28,526 | ₹58,479 |

## 3. First-year compliance calendar (to confirm with a CA/CS)

| When | Filing / task |
|---|---|
| Before M1 | Incorporate Pvt Ltd (SPICe+), PAN/TAN, bank account; founders' agreement and inventor assignments |
| Within 30 days | Appoint first auditor (ADT-1); first board meeting |
| Within 180 days | Commencement of business (INC-20A) |
| On turnover/inter-state supply | GST registration; then GSTR-1 (11th) and GSTR-3B (20th) monthly, or QRMP quarterly |
| Quarterly | TDS returns; board meeting (at least 4 a year) |
| M2 | Provisional patent filing (planned); trademark applications |
| M4 | CDSCO classification opinion; ethics committee approval for the user study |
| M9–M10 | CDSCO registration/licence application for TremoSense (class to be confirmed) |
| 30 Sep | DIR-3 KYC for directors; AGM within six months of year end |
| 31 Oct / Nov | ITR-6; AOC-4 and MGT-7 annual ROC filings |

| When | Filing / task |
|---|---|
| Any time | DPIIT Startup recognition; Startup India Seed Fund application (applied for, not secured) |

## 4. Founder-agreement clauses (proposed)

1. **Equity and vesting:** founder shares vest over 4 years with a 1-year cliff; unvested shares return to the company on departure.
2. **Roles:** Pranshu — CEO, wearable hardware and firmware; Pawan — control, gimbal and power electronics. Both remain the hardware and test engineers.
3. **Decision-making:** day-to-day decisions by the role owner; fundraising, new products, hiring above ₹30,000/month and any medical claim need both founders.
4. **Departure:** good-leaver keeps vested shares; bad-leaver (fraud, breach) sells vested shares back at face value; 3-month notice and handover.
5. **IP assignment:** all IP created for AVERRA (including pre-incorporation prototypes) is assigned to the company before any patent filing.

M4 · Activity 1 · Mandatory Model M4.1

# Team & Incentive Blueprint

| | | | |
|---|---|---|---|
| **Module** 4 | **Activity** 1 | **Date completed** 3 Oct 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · simulated mock interview | | | |
| *Data note: Org chart and ESOP are from the model; the mock-interview record is simulated.* | | | |

## 1. 12-month org chart and hiring sequence

| Month | Team | Why then |
|---|---|---|
| M1–M6 | Pranshu Kumar (CEO, wearable hardware & firmware) · Pawan Shinde (control, gimbal & power) — both hardware and test engineers | Phase 1 is engineering and validation; no hires |
| M7–M12 | Founders + contract support (regulatory consultant, industrial designer, contract assembler) | Regulatory filing and pilot batch; contracts, not payroll |
| M19 | + Clinic-network sales lead · + Customer support & operations | After first sales prove demand |
| M25 | + App & firmware engineer (ParkinSense) | ParkinSense wellness launch |

The plan's first hires fall after month 12, so the 12-month chart is the two founders plus contractors (Forecast sheet hiring lines).

## 2. Role descriptions

| Role | Responsibilities | What good looks like |
|---|---|---|
| Clinic-network sales lead (M19) | Sign and support neurology and OT clinics; run in-clinic trials; manage clinic margin terms | 30 active clinics by M30; ≥4 TremoSense sales per clinic per month; CAC ≤ ₹1,079 |
| Customer support & operations (M19) | Onboarding calls, 7-day check-ins, returns, dispatch, stock | Onboarding within 48 h; ≥70% daily use at day 30; return rate <5% |
| App & firmware engineer (M25) | ParkinSense mobile app, BLE firmware, cloud sync | App store release by M27; crash-free sessions ≥99%; no data-privacy incidents |

## 3. Structured mock-interview guide (same questions for every candidate)

1. Tell me about a time you got a clinician or institution to adopt something new. What exactly did you do?
2. Walk me through how you tracked a sales pipeline last time. What numbers did you watch?
3. Describe a deal you lost. What did you learn?
4. A caregiver says the device isn't helping after a week. What do you do in the first call?
5. Score: 1–5 on evidence of past behaviour, clarity, customer empathy, numeracy.

| Interviewer (classmate) | Date | Q1 | Q2 | Q3 | Q4 | Total /20 | Notes |
|---|---|---|---|---|---|---|---|
| Classmate A as candidate for clinic sales lead (simulated record) | 3 Oct 2026 | 4 | 3 | 4 | 5 | 16 | Strong empathy and concrete examples; weaker on pipeline numbers |

> Mock-interview record above is simulated; the class exercise uses the same four questions and scoring sheet.

## 4. Founder equity-split method

| Factor (weight) | Pranshu (1–10) | Pawan (1–10) | Weighted P / W |
|---|---|---|---|
| Idea and IP (25%) | 8 | 7 | 2.00 / 1.75 |
| Capital contributed (15%) | 5 | 5 | 0.75 / 0.75 |
| Full-time commitment (35%) | 9 | 9 | 3.15 / 3.15 |
| Prior expertise (25%) | 8 | 8 | 2.00 / 2.00 |
| **Total** | | | **7.90 / 7.65** |

Founder share = Σ(weight × score) for that founder ÷ Σ(weight × score) for both founders × 82.71% (founders' total after Round 1 with a 5% ESOP pool; Funding Scenarios sheet). Pranshu 7.90 ÷ 15.55 = 50.8% of founder equity → 42.02% after

Round 1; Pawan 7.65 ÷ 15.55 = 49.2% → 40.69%. Vesting: 4 years, 1-year cliff, monthly thereafter.

## 5. ESOP pool and a sample grant

The model reserves a **5% ESOP pool**, enough for the three planned hires (≈1–1.5% each) plus a reserve. The course suggests 10–15%; we keep 5% now to limit dilution in a small round and plan to top it up at tranche 2 when the team grows (flagged for the founders to decide).

Sample grant: 0.5% × post-money ₹85,51,171 = **₹42,756** on paper at the proposed valuation (illustrative; not a market value).

M4 · Activity 2 · Mandatory Model M4.2

# Growth-Engine Dashboard

| | | | |
|---|---|---|---|
| **Module** 4 | **Activity** 2 | **Date completed** 7 Oct 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · simulated campaign data | | | |
| *Data note: Campaign, funnel and cohort figures are simulated; ICE scores are the founders' judgement.* | | | |

Simulated semester data: one ₹500 paid boost and one organic post (September 2026), and ParkinSense prototype-beta cohorts (July–September 2026).

## 1. Campaign tracker

| Channel | Spend ₹ | Reach | Clicks | Enquiries | Conversions | CAC | ROAS |
|---|---|---|---|---|---|---|---|
| Instagram boost of the bench video (Mumbai + Pune, 25–65) | ₹500 | 6,200 | 184 (3.0%) | 23 | 3 refundable deposits | ₹500 ÷ 3 = ₹167 | ₹3,000 ÷ ₹500 = 6.0× (upper bound; deposits are refundable) |
| Support-group WhatsApp post (organic) | ₹0 | 350 | 41 | 11 | 2 pilot sign-ups | ₹0 | — |
| **Total** | **₹500** | **6,550** | **225** | **34** | **3 deposits · 5 pilot users** | **₹167 per deposit** | |

The ₹167 campaign CAC is below the model's planned TremoSense CAC of ₹1,079, but it covers refundable deposits, not completed sales, so the model CAC is kept unchanged.

## 2. Cohort retention

| Cohort (joined) | Users | Active Month 1 | Active Month 2 | Active Month 3 |
|---|---|---|---|---|
| Jul 2026 (prototype beta) | 8 | 8 (100%) | 6 (75%) | 5 (63%) |
| Aug 2026 | 10 | 10 (100%) | 8 (80%) | 7 (70%, Oct to date) |
| Sep 2026 | 12 | 12 (100%) | 9 (75%, Oct to date) | — |

## 3. Growth experiment backlog, ICE-scored (scores are the founders' current judgement)

| Experiment | Pass mark | Impact | Confidence | Ease | ICE total | Rank |
|---|---|---|---|---|---|---|
| In-clinic trial → purchase (10 clinics, 8 weeks) | ≥30% buy at ₹9,999; CAC ≤ ₹1,079 | 9 | 5 | 5 | 19 | 1 |
| ParkinSense wellness beta (100 users, 8 weeks) | ≥40% weekly active; ≥25% would subscribe | 7 | 5 | 6 | 18 | 2 |
| Support-group referrals (PD/ET caregiver groups) | ≥15% refer; CAC ≤ ₹800 | 6 | 4 | 7 | 17 | 3 |

ICE total = Impact + Confidence + Ease (each 1–10). Rank 1 runs first because it tests the riskiest commercial assumption (willingness to pay through clinics).

M4 · Activity 3 · Mandatory Model M4.3

# Fundraising Model, Cap Table & Investor Pitch

| | | | |
|---|---|---|---|
| **Module** 4 | **Activity** 3 | **Date completed** 10 Oct 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · model figures | | | |
| *Data note: Ask, valuation and cap table are from the model; ₹75 L pre-money is a proposed negotiation target.* | | | |

## 1. The ask and the milestone it funds

Raise **₹10,51,171** (Round 1) to fund Milestone 1 (M1–M6): both founders full-time, 12 study units, the preregistered user study, a provisional patent filing (planned) and a CDSCO classification opinion. Amount = Phase-1 peak cash gap ₹8,75,976 × 1.2 buffer.

## 2. Valuation

**Proposed pre-money: ₹75,00,000 — a working negotiation target, not an independently verified market valuation.** Methods used, named explicitly:

- **Cost-to-duplicate (floor):** 34 person-months × ₹1,25,000 + ₹2,25,000 materials = ₹44,75,000.
- **Berkus-style scorecard (cross-check):** ₹15.0 L + ₹20.0 L + ₹12.5 L + ₹10.0 L + ₹2.5 L = ₹60,00,000.
- The premium above the cross-check reflects two products and a clinical evaluation in progress (founder-reported).

## 3. Arithmetic

| | Formula | Value |
|---|---|---|
| Post-money | Pre-money + investment | ₹75,00,000 + ₹10,51,171 = **₹85,51,171** |
| Investor ownership | Investment ÷ post-money × 100 | ₹10,51,171 ÷ ₹85,51,171 = **12.29%** |
| ESOP pool | set aside | **5%** |
| Founders | 100% − investor − ESOP | **82.71%** |

## 4. Cap table

| Holder | Before Round 1 | After Round 1 (at ₹75 L pre) |
|---|---|---|
| Founders | 100% | 82.71% |
| ESOP pool | — | 5.00% |
| Round-1 investor | — | 12.29% |
| **Total** | 100% | 100% |

The model treats the 5% ESOP pool as created at Round 1 alongside the investor (founders = 100% − 12.29% − 5%). Check: 82.71% + 5.00% + 12.29% = 100.00%.

## 5. Use of funds (Phase 1, M1–M6)

| Category | Amount | Share | What it buys |
|---|---|---|---|
| Team | ₹2,70,000 | 31% | Both founders full-time for six months |
| Validation (both products) | ₹1,58,341 | 18% | 12 TremoSense study units and the user study (n ≥ 10); ParkinSense reference testing |
| Regulatory & quality | ₹30,000 | 3% | CDSCO classification opinion and ethics approval |
| Prototyping, tools, IP | ₹2,53,000 | 29% | Provisional patent, test equipment and prototype iterations |
| Go-to-market & admin | ₹85,000 | 10% | Clinic outreach for the trial-to-purchase test; company admin |
| Contingency | ₹79,634 | 9% | Buffer for slips and price changes |
| **Total Phase 1** | **₹8,75,976** | 100% | Round = this × 1.2 = ₹10,51,171 |

**Runway:** 7.2 months at Phase-1 burn. **After this round:** tranche 2 of ₹37.71 L must close by M7 (Jan 2028), priced on the M6 result; total to break-even ₹48.22 L. **Startup India Seed Fund grant (₹20 L): applied for, not secured, and not counted.**

## 6. Short investor pitch

> AVERRA engineers better movement intelligence. TremoSense is an active stabilising spoon that rejects about 90% of tremor on our bench, founder-reported, at ₹9,999, a quarter of imported alternatives; ParkinSense is a ₹7,999 wellness band with tremor detection, in a clinical evaluation with ICMR. Every unit is profitable from the first sale: ₹2,538 contribution on TremoSense. We are raising ₹10.51 L at a ₹75 L pre-money for 12.3% to reach one clear answer in six months: a preregistered user study with at least ten people. If the median tremor reduction is below 30%, we stop TremoSense.

Mock-panel questions (simulated rehearsal, 14 Oct 2026). Questions: “Why ₹75 L and not ₹60 L?” → cost-to-duplicate floor plus two products and an ongoing clinical evaluation; “What if the study fails?” → preregistered kill rule at <30%; “Why clinics, not online?” → caregivers want to try before buying.

M5 · Activity 1 · Mandatory Model M5.1

# Sustainable & Ethical Enterprise Blueprint

| | | | |
|---|---|---|---|
| **Module** 5 | **Activity** 1 | **Date completed** 12 Oct 2026 | **Venture** AVERRA |
| **Evidence status:** Completed | | | |
| *Data note: Policy commitments; no external certification is claimed.* | | | |

## 1. Stakeholders beyond customers and investors

| Stakeholder | How AVERRA affects them | Our commitment |
|---|---|---|
| Caregivers and families | Time and stress at meals; they often pay | Honest claims; rental option; clear return policy |
| Clinicians and OT therapists | Referrals carry their reputation | Share study results, including negative ones; no referral kickbacks |
| Employees and contractors | Assembly and test work, data access | Fair pay; safety training for battery handling; data-access controls |
| Suppliers and contract assembler | Volume, payment terms | Pay within 30 days; no forced price cuts after orders |
| Environment | LiPo batteries, electronics, plastic | Take-back and refurbishment programme; RoHS-compliant parts |
| Community / research participants | Study participants' health data | Ethics approval, informed consent, data minimisation |

## 2. One specific ESG improvement

**Device take-back and refurbishment (environmental + social).** Every TremoSense and ParkinSense comes with a prepaid return label at end of life or after a rental ends. Returned units are tested on the same calibration rig, refurbished and re-rented at a lower rate to families who cannot afford ₹9,999; LiPo cells and e-waste go to an authorised recycler under India's E-Waste (Management) Rules 2022. Target: ≥50% of rental units refurbished at least once.

## 3. Ethical red lines and scenario tests

| Red line | Tempting, realistic scenario | What we do |
|---|---|---|
| We will never claim ParkinSense diagnoses or predicts Parkinson's until clinical results support it | A marketing partner says “early Parkinson's detection” would triple sign-ups | Refuse. Market it as a wellness band with tremor insight; add clinical claims only after ICMR results and regulatory approval |
| We will never sell or share identifiable health or movement data | An insurer offers ₹10 L a year for anonymised tremor data | Decline unless users give explicit, separate opt-in consent and the data is genuinely de-identified under India's DPDP Act 2023 |
| We will never hide or reword a failed validation result | The M6 study shows 35% reduction, below our 50% target, just before tranche 2 | Report the result as preregistered, follow the redesign rule, and tell investors before they decide |

## 4. Responsibilities specific to biomedical wearables and assistive technology

- **Data privacy:** minimal data collection; on-device or encrypted transfer; deletion on request (DPDP Act 2023).
- **Accessibility:** large controls, no small screens to read, caregiver-friendly onboarding, regional-language manuals, rental for affordability.
- **Truthful claims:** “≈90% on a bench simulator (founder-reported)” is never shortened to “90% less tremor”.
- **Unvalidated features:** HR/HRV/$SpO_2$ are labelled research-grade until validated against reference devices.

M5 · Activity 2 · Mandatory Model M5.2

# Live Launch & Iteration Report

| | | | |
|---|---|---|---|
| **Module** 5 | **Activity** 2 | **Date completed** 14 Oct 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · simulated pilot launch | | | |
| *Data note: Pilot feedback and iteration results are a simulated case study consistent with M2.2 and M4.2.* | | | |

> Simulated case study: a 4-week semester pilot with 5 users of the current prototype across two clinics, 15 Sep – 12 Oct 2026, run on the M2.2 protocol. The funded launch plan follows in section 5.

## 1. Pilot-launch plan

| Item | Plan |
|---|---|
| What | Semester pilot: 5 users, current prototype, supervised meals (Sep–Oct 2026). Funded pilot: 12 study units after ethics approval (M5–M6 of the plan) |
| Who | Semester: 5 adults with tremor and their caregivers via 2 clinics. Funded: ≥10 via 2–3 neurology/OT clinics |
| How long | 4 weeks |
| Feedback channels | Session interview, 7-day WhatsApp check-in, short exit questionnaire |
| Decision rule | Same preregistered thresholds as M2.2 |

## 2. Evidence template

| # | Date | User type | Feedback (simulated) | Theme |
|---|---|---|---|---|
| 1 | 21 Sep 2026 | Man, 69, PD | “The food stays on the spoon now. But my hand gets tired holding it.” | Weight |
| 2 | 22 Sep 2026 | Caregiver, daughter | “He finished his whole plate without me helping. First time in months.” | Independence |
| 3 | 24 Sep 2026 | Woman, 66, ET | “It is a little heavy, and I wasn't sure when it was switched on.” | Weight; status light |
| 4 | 26 Sep 2026 | OT therapist | “Patients need a clear on/off light and a lighter handle.” | Weight; status light |
| 5 | 28 Sep 2026 | Man, 72, PD | “Charging every night is fine if it lasts through dinner.” | Battery |

| Recurring theme | How many users | Change we made | What happened after |
|---|---|---|---|
| Handle too heavy | 3 of 5 | Moved the battery lower in the handle and switched to a lighter boost board: 182 g → 164 g | Re-test with 4 of the 5 users on 5 Oct: 3 of 4 preferred the new handle; none reported tiredness in a 15-minute meal |
| Unclear on/off state | 2 of 5 | Added a green status LED | No more confusion reported |

## 3. Crisis and contingency plan

| Risk | Likelihood | Impact | Response |
|---|---|---|---|
| A servo or battery failure during a meal (spill, burn risk from food, or overheating) | Medium during the pilot | High (user safety, clinic trust) | Mechanical end-stops limit head travel; thermal cut-off in firmware; supervised first use; stop-use notice and replacement within 48 h; incident log reviewed with the clinic; pause the pilot if two incidents occur |
| Tranche 2 closes late | Medium | High (zero-cash M7) | Pause hiring, inventory and testing; defer stipends; buffer lasts to ≈M14 |

## 4. Continue / kill call

**CONTINUE.** In the simulated pilot, users needed less help at meals, the main complaint (weight) was fixed and re-tested, and 3 of 5 caregivers left refundable deposits. The decision is provisional until the funded n ≥ 10 study.

## 5. Funded launch plan (from the financial model)

| When | Launch step | Budget (model) | KPI |
|---|---|---|---|
| M6 (Dec 2027) | Preregistered user study, n ≥ 10 | ₹1.00 L study + ₹0.58 L units | Median reduction ≥50%; ≥60% would pay |
| M9–M11 | EMC pre-compliance, CDSCO filing, 100-unit pilot batch | Regulatory ₹6.30 L (Phase 2) | Filing submitted; batch yield ≥95% |
| M12 (Jun 2028) | First TremoSense sales through clinics | Outreach ₹3.15 L (Phase 2) | 15 units in Year 1; CAC ≤ ₹1,079 |
| M19–M23 | Clinic network to 30 clinics; price to ₹8,999 after verified cost cut | Sales lead + support hires | Break-even M23 (May 2029) |
| M25 | ParkinSense wellness launch online | Inventory + D2C CAC ₹1,500 | 1,380 units in Year 3 |

## 6. Channel plan

| Channel | Role | Contribution per TremoSense (model) |
|---|---|---|
| Clinic / OT referral | Main channel: try before buying, clinician trust | ₹2,538 (15% clinic margin) |
| Direct online | Repeat buyers, referrals, ParkinSense | ₹3,966 |
| Institutional procurement | Hospitals and NGOs; 60-day terms | ₹3,014 |
| Rental | Families who cannot pay upfront | ₹576/month net; device recovered in 9.3 months |

## 7. Risk register

| Risk | Likelihood | Impact | Mitigation |
|---|---|---|---|
| Bench results do not transfer to real meals | Medium | High | Preregistered study; kill below 30% median reduction |
| Families will not pay ₹9,999 | Medium | High | Rental at ₹799/month; clinic trial-to-purchase test |
| CDSCO classification or timeline slips | Medium | Medium | Classification opinion at M4; consultant on contract |
| Component cost stays at Scenario B | Medium | Medium | Hold ₹9,999 until Scenario C cost is verified |
| Tranche 2 late (zero cash M7) | Medium | High | Pause hiring and inventory; buffer to ≈M14 |
| Data-privacy incident (ParkinSense) | Low | High | Encryption, minimal data, DPDP Act compliance |

M5 · Activity 3 · Mandatory Model M5.3

# Full Venture Launch & Fundraising Package

| | | | |
|---|---|---|---|
| **Module** 5 | **Activity** 3 | **Date completed** 16 Oct 2026 | **Venture** AVERRA |
| **Evidence status:** Completed · thesis draft | | | |
| *Data note: Summarises M1.1–M5.2; simulated items keep their labels.* | | | |

## 1. Package contents

| Component | Where | Status |
|---|---|---|
| Pitch deck (17 slides) | deck/AVERRA_Pitch.pptx and averraproject.vercel.app | Done |
| Financial model | model/AVERRA_Financial_Model.xlsx | Done (forecast, not actuals) |
| Cap table and fundraising model | M4.3; Funding Scenarios sheet | Done |
| Customer evidence | M1.1 | Simulated case study (M1.1); replace with interview records |
| MVP validation | M2.2 | Bench evidence + simulated 5-user pilot; funded study planned |
| Launch evidence | M5.2 | Simulated pilot (M5.2) |

## 2. Why this could fail (single biggest, specific risk)

**Bench rejection may not turn into easier eating. Real food, real spoons and tired hands are harder than a motorised simulator.**

**Stop or change direction if:** the M6 study (n ≥ 10) shows median tremor reduction below 30% → stop TremoSense and focus on ParkinSense; 30–50% → one redesign and re-test; it works but fewer than 60% would pay → pivot to clinic rental at ₹799/month; tranche 2 does not close by M7 → pause spending and extend to ≈M14 on the buffer.

## 3. Investment memo

**Company.** AVERRA, biomedical technology. “Engineering better movement intelligence.” Founders: Pranshu Kumar and Pawan Shinde (both hardware and test engineers).

**Problem.** About 4.6 lakh Indians aged 60+ live with Parkinson's and ≈12.8 lakh have essential tremor (single-study estimate). Movement is measured only in short clinic visits, and tremor makes eating hard and embarrassing (published studies; simulated interview case study in M1.1).

**Products and status.** TremoSense, an active stabilising spoon: ≈90% tremor rejection on a bench simulator (founder-reported; logs to be published); 2 informal user trials. ParkinSense, a ₹7,999 screenless wellness band with tremor detection: steps, tremor and rest-vs-movement bench-tested; heart metrics need validation; clinical evaluation with ICMR in progress, results pending. No diagnostic claims.

**Market.** Bottom-up TAM ₹416.9 Cr, SAM ₹51.9 Cr, Year-3 SOM ₹109.0 L; general-wearable upside shown separately.

**Business model and unit economics.** Clinic-referred device sales and rental; ParkinSense online with an optional plan; RUO kits for labs. TremoSense contribution ₹2,538 (31.4%) at ₹9,999; LTV:CAC 3.3×; CAC paid back on the first sale.

**Financials (forecast).** Year-1 costs ₹31.57 L; break-even M23 · May-29; Year-3 operating profit ₹18.93 L on ₹2.12 Cr cash in; total to break-even ₹48.22 L.

**The ask.** ₹10,51,171 at a proposed ₹75,00,000 pre-money (negotiation target; cost-to-duplicate floor ₹44.75 L, Berkus cross-check ₹60 L) = 12.29% to the investor; founders 82.71% after a 5% ESOP pool. Funds Milestone 1: the preregistered user study by M6. Tranche 2 ₹37.71 L by M7. Startup India grant applied for, not secured.

**Key risk and kill rule.** Bench results may not transfer to real meals; stop TremoSense below 30% median reduction.

## 4. Personal entrepreneurial thesis

| Question | Draft answer |
|---|---|
| Start, operate, fund or join? | START: I would start this venture. |
| Evidence from my M1–M5 work | 5 of 6 interviewees described weekly eating problems and 4 had already paid for a workaround (M1.1, simulated); the 5-user pilot reached 58% median reduction, indicative only (M2.2); contribution is ₹2,538 per TremoSense at ₹9,999 and LTV:CAC 3.3× (M2.3, M3.2). |
| What changed my thinking | Caregivers, not patients, make the buying decision, so clinics and caregivers became our channel; weight mattered more than I expected. |
| What would change my answer | A larger study below 30% median reduction, or fewer than 30% of trial users buying at ₹9,999. |

# References

Prices and figures were checked on the dates stated in each label (most competitor prices on 16 Oct 2026; FX on 12 Oct 2026). Accessed for this portfolio: 16 Oct 2026.


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   https://pmc.ncbi.nlm.nih.gov/articles/PMC11890171
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    https://9to5mac.com/2022/06/13/apple-watch-parkinsons/
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16. Garmin Cirqa India launch, ₹25,990 (Business Today, 7 Oct 2026)
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https://averraproject.vercel.app

# Evidence appendix

## Evidence status of all 15 models

| Model | Title | Status |
|---|---|---|
| M1.1 | Customer-Discovery Evidence Pack | Completed · simulated interview case study |
| M1.2 | Opportunity Scorecard & “Why Now?” Memo | Completed · sourced and model figures |
| M1.3 | Venture Money-Model & First-Budget Pack | Completed · model figures |
| M2.1 | Evidence-Backed Business Model Canvas + Value Proposition Canvas | Completed |
| M2.2 | MVP Validation Experiment & Decision | Completed · simulated semester pilot; funded study planned |
| M2.3 | Pricing, Margin & Funnel Performance Pack | Completed · model pricing; simulated funnel |
| M3.1 | Operating System & Delivery-Process Blueprint | Completed · estimated timings |
| M3.2 | Unit-Economics & Runway Model | Completed · model forecast |
| M3.3 | Legal-Structure, GST & Compliance Blueprint | Completed · illustrative GST working |
| M4.1 | Team & Incentive Blueprint | Completed · simulated mock interview |
| M4.2 | Growth-Engine Dashboard | Completed · simulated campaign data |
| M4.3 | Fundraising Model, Cap Table & Investor Pitch | Completed · model figures |
| M5.1 | Sustainable & Ethical Enterprise Blueprint | Completed |
| M5.2 | Live Launch & Iteration Report | Completed · simulated pilot launch |
| M5.3 | Full Venture Launch & Fundraising Package | Completed · thesis draft |

## Evidence already in hand

- Two working prototypes and repositories (ParkinSense develop branch; TremoSense project).
- Bench results: ParkinSense 8/8 bench conditions, ≈7 mA; TremoSense ≈90% bench rejection (founder-reported, logs to be published).
- Dated, sourced market and competitor figures (References).
- Financial model with live formulas (11 sheets).
- ParkinSense dashboard screenshot reading tremor at 4.88 Hz (pitch website).

## Real evidence still to collect (replaces simulated entries)

| Model | Item still to be evidenced | How to collect it | Done |
|---|---|---|---|
| M1.1 | Interview records (5+), consent and verbatim notes | PD/ET patients and caregivers via clinics and support groups | ☐ |
| M2.2 | Funded study results (n ≥ 10); published bench logs for the ≈90% figure | After ethics approval; publish logs in the repository | ☐ |
| M2.3 / M4.2 | Campaign dashboard export; deposit receipts | Run the ₹500 boost; UPI receipts | ☐ |
| M4.1 | Signed founders' agreement with agreed scores | Founders' meeting; CA/lawyer review | ☐ |
| M5.2 | Pilot feedback records and the handle re-test log | Session notes; WhatsApp check-ins | ☐ |
| All | ICMR clinical-evaluation results (ParkinSense) | Add when published; no clinical claim before then | ☐ |
| All | Provisional patent filing receipt (filing planned) | File and attach the receipt | ☐ |

# Consistency review

Cross-checks between this portfolio, the pitch deck, the website and the financial model.

| # | Item | Resolution in this portfolio | Status |
|---|---|---|---|
| 1 | Course asks for a 10–14 slide pitch; deck and website have 17 | 14-slide timed version: merge Why now into Problem, move Architecture to backup, fold Advantage into the close | Open (deck) |
| 2 | Problem slide quotes published research, not own interviews | M1.1 adds interview records; deck to add 2–3 quotes once real records exist | Open (deck) |
| 3 | ESOP pool 5% vs course example 10–15% | Kept at 5% to limit dilution in a small round; top-up planned at tranche 2 (M4.1, M4.3) | Resolved |
| 4 | Milestones listed M25 before M23 | Shown chronologically in M5.2 launch plan | Resolved here |
| 5 | Semester pilot (2026) vs funded study at M6 (Dec 2027) | Pilot is n = 5 and indicative only; the funded study keeps the kill / continue decision (M2.2, M5.2) | Resolved |
| 6 | Campaign CAC ₹167 vs model CAC ₹1,079 | Campaign counts refundable deposits, not sales; model CAC unchanged (M4.2) | Resolved |
| 7 | Bad-growth scenario not a workbook toggle | Computed in M3.2 from the Price Matrix | Resolved here |

# Final completeness checklist

| Check | Result |
|---|---|
| All 15 mandatory models present, labelled M1.1–M5.3, in module order | Yes |
| Each sheet has module, activity, title, date, evidence status and data note | Yes |
| Figures cross-checked against the financial model | Yes — market, pricing, unit economics, forecast and cap table generated from the model |
| Funding ask, valuation, ownership and ESOP | ₹10,51,171; ₹75 L pre-money (proposed negotiation target); post ₹85,51,171; investor 12.29%; ESOP 5%; founders 82.71% |
| Grant shown as secured anywhere? | No — Startup India Seed Fund applied for, not secured |
| Patent or clinical claims stated as fact? | No — provisional filing planned; ICMR results pending; no diagnostic claims |
| Simulated data labelled | Yes — status line and data note on M1.1, M2.2, M2.3, M4.1, M4.2, M5.2 |
| Placeholders or blank fields left | None, except the real-evidence checklist boxes to tick on collection |

# Submission checklist (physical copy)

1. Print single-sided on A4, colour if possible (status labels are colour-coded but also written in words).
2. Fill every date field and sign/initial each sheet.
3. Complete the pending items you have evidence for; leave the rest visibly marked pending.
4. Attach evidence behind the matching sheet: interview notes, price-test responses, receipts/preorder proof, campaign screenshots, mock-interview notes.
5. Order: cover → contents → M1.1 … M5.3 → references → evidence appendix → inconsistencies → this checklist.
6. Bind with a clear cover; add tabs for Modules 1–5 so pages can be opened quickly in the viva.
7. Bring the deck on a USB drive and the website link as backup.
8. Before the viva, re-read M1.2 (TAM/SAM/SOM), M3.2 (runway, zero-cash date) and M4.3 (valuation arithmetic).